

Non-Gaussian Market Returns: Evidence from Normal and Student-t Models

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I. Introduction

A common, simplistic assumption in finance is that daily asset returns are approximately Gaussian. By definition, under this assumption, large negative moves are extremely rare. Yet, in practice, markets appear to experience ‘crashes’ more often than a Normal model would predict, suggesting heavier or ‘fatter’ tails than the Gaussian benchmark (Cont 2001). This motivates the basic question for this project: *for typical equity index returns, does a fat-tailed distribution provide a statistically meaningful, better likelihood fit than the Normal model, and what does this imply for risk measurement?*

This project tests Normality against a heavy-tailed alternative using maximum likelihood estimation (MLE). We estimate a Normal model and a Student-t model for daily log returns, and compare in-sample likelihood and information criteria (AIC and BIC). Because the Normal distribution is the limiting case of the Student-t as degrees of freedom tend to infinity, the Student-t provides a natural nested generalization of the Gaussian benchmark. The core empirical focus in the code implementation presented here is QQQ, an ETF tracking the Nasdaq-100, with a training and testing split designed to evaluate out-of-sample fit.

II. Background

Empirically, return distributions often display excess kurtosis (degree of ‘tailedness’) relative to a Normal distribution (Cont 2001). A Student-t distribution is a standard parametrization used to model this behavior: it resembles a Normal distribution in the center but assigns greater probability mass to tail events when degrees of freedom are small. In this setting, the degrees of freedom parameter controls tail thickness, with lower values implying heavier tails. The Normal distribution is recovered in the limit as degrees of freedom increases without bound, which makes it a convenient baseline for model comparison.

From an estimation perspective, MLE provides a direct way to compare models using their maximized log-likelihoods, and information criteria such as AIC and BIC penalize additional parameters (measuring goodness-of-fit against model complexity). Since the Student-t adds one parameter (degrees of freedom) relative to the Normal, a higher in-sample log-likelihood is not automatically telling of its efficacy. This is why AIC and BIC comparisons are useful complements, and why an explicit train-test split is included.

III. Data and Methodology

A. Data and pre-processing

We use daily equity return data obtained through the Python package [yfinance](#), which provides programmatic access to historical price data sourced from Yahoo Finance. While Yahoo Finance data are not intended for professional trading or regulatory use, they are widely used in academic settings and are appropriate for the scope of this project.

Our primary dataset consists of daily prices for five well-known and long-standing U.S. securities: SPY, QQQ, AAPL, JPM, and XOM, covering the period from January 1, 2010 through December 17, 2025. The core empirical analysis in this paper focuses on QQQ, an exchange-traded fund tracking the Nasdaq-100 index, which provides a liquid and representative sample of large-cap growth-oriented equities.

For each asset, we use the Adjusted Close price series. According to Yahoo Finance, the adjusted close reflects the closing price after accounting for stock splits and dividend distributions, using appropriate adjustment multipliers consistent with Center for Research in Security Prices (CRSP) standards. Using adjusted prices ensures that computed returns reflect true economic gains and losses rather than mechanical price changes due to corporate actions.

Returns are computed as daily log returns, defined as:

$$r_t = \log(P_t) - \log(P_{t-1})$$

where P_t denotes the adjusted closing price on day t . Log returns are standard in financial econometrics because they are approximately additive over time and naturally arise under continuous-time models of asset prices (Tsay 2010). It follows that we intuit the data of changes in price follows a log normal distribution.

The data are split into a training sample and a test sample using a chronological split to preserve the forecasting interpretation of risk measures. The training period runs from January 5, 2010 through October 5, 2022 (3,211 observations), while the test period spans October 6, 2022 through December 17, 2025 (803 observations). A time-based split is essential because financial returns exhibit volatility clustering and structural changes over time; random splitting would introduce look-ahead bias and invalidate out-of-sample evaluation.

Note: All analysis was conducted in Python.

B. Methodology

Distributional Models

We model daily log returns using two parametric distributions:

1. *Normal distribution*

$$r_t \sim N(\mu, \sigma^2),$$

which represents the standard Gaussian assumption commonly used in finance.

2. *Student-t distribution*

$$r_t \sim t_v(\mu, \sigma),$$

Where:

- a. v denotes the degrees of freedom parameter, controlling tail thickness. Lower values of v imply heavier tails. It is important to note here that the Normal distribution is recovered when $v \rightarrow \infty$, making the Student-t distribution a natural generalization of the Gaussian model.

Both models are estimated using maximum likelihood estimation (MLE) on the training sample. For the Normal model, the MLEs of the mean and variance have closed-form expressions. For the Student-t model, estimation proceeds by fixing the degrees of freedom v on a grid from 1 to 150 and, for each value of v , numerically maximizing the likelihood with respect to μ and σ . The value of v that maximizes the log-likelihood is selected as the optimal degree of freedom.

Model Comparison

Models are compared using multiple complementary criteria:

- In-sample log-likelihood, which measures goodness-of-fit.
- Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), which penalize model complexity and allow fair comparison between the two-parameter Normal model and the three-parameter Student-t model.

Out-of-sample log-likelihood, evaluated on the test sample using parameters estimated from the training data.

- Although a likelihood ratio statistic is reported for comparison, formal chi-square inference is only approximate because the Normal model lies on the boundary of the Student-t parameter space (as $v \rightarrow \infty$). For this reason, AIC, BIC, and out-of-sample performance are emphasized as more reliable criteria.

Risk Measurement and Backtesting

To assess the economic implications of distributional assumptions, we compute Value at Risk (VaR) under both models. For a given confidence level α , the one-day VaR is defined as the α -quantile of the fitted return distribution. VaR is evaluated at the 1% and 5% levels using parameters estimated from the training sample.

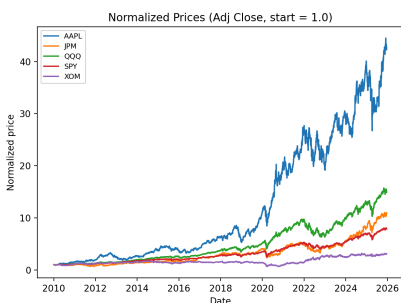
Out-of-sample VaR accuracy is assessed on the test sample using standard backtesting procedures:

- Kupiec's unconditional coverage test, which evaluates whether the observed frequency of VaR violations matches the nominal level.
- Christoffersen's independence test, which assesses whether violations occur independently over time.
- Christoffersen's conditional coverage test, which jointly tests correct coverage and independence.

IV. Results and Analysis

This section presents the empirical results from comparing Gaussian and Student-t models for daily log returns, focusing on QQQ. We begin with exploratory diagnostics, proceed to likelihood-based estimation and model comparison, and conclude with out-of-sample risk evaluation using Value at Risk (VaR).

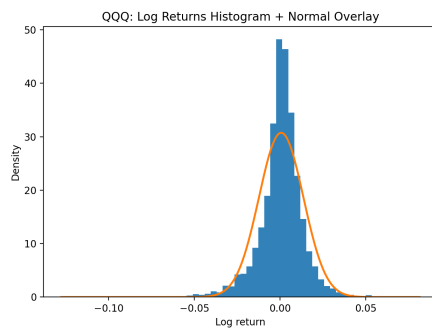
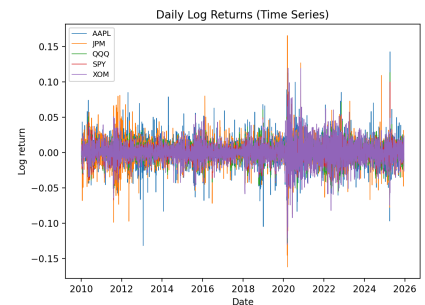
Exploratory Diagnostics



First our choice of QQQ was based on the normalized returns over the course of the data. QQQ is an ETF following the NASDAQ index, making it an apt representation of the market. Additionally, it was fairly

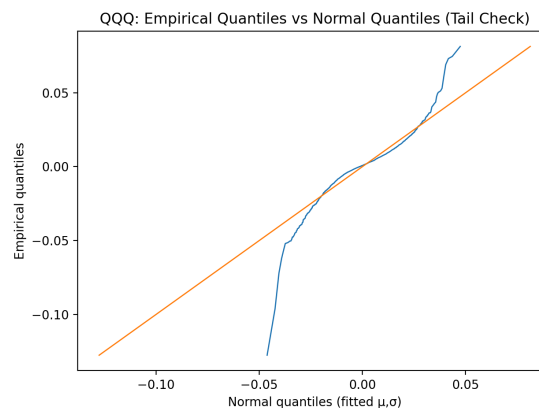
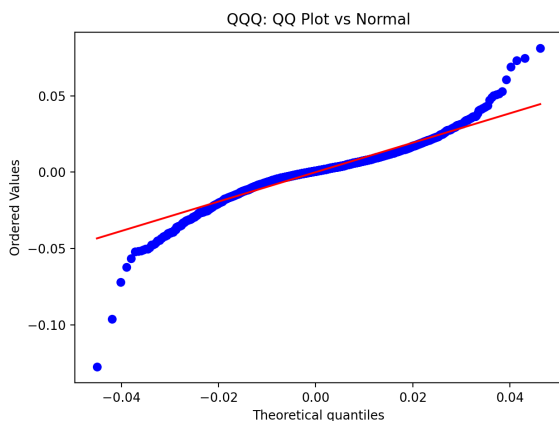
moderate between volatile tech stocks and less volatile oil and gas type stocks.

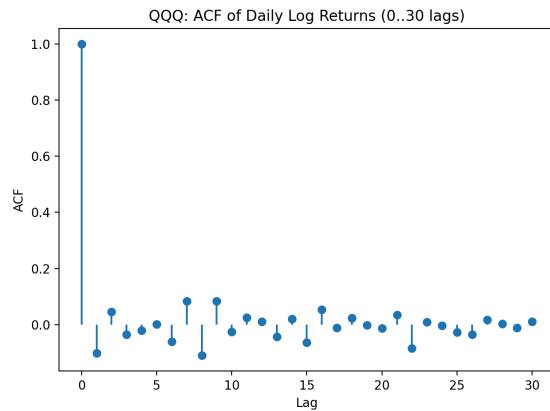
A critical issue in the data is the heteroskedasticity (seen clearly in the figure on the right), since the volatility modeled across time suggests non homogenous variance(Tsay 2010, Bollerslev 1987).



To begin with, looking at the fitted normal overlay on the histogram of QQQ data (left) shows us that the normal pdf underestimates the concentration at both the center and the tails of the distribution, suggesting that the better choice should have fatter tails.

The QQ plots (below) reveal that, along the extremes of the dataset, the data deviates from the expectations of the fitted normal distribution.





Note: the auto correlation function (visualized on the left) was used to look at whether prior returns across different time windows, from 1 to 30, were correlated with the current day's returns. The ACF of returns is close to zero at short lags, suggesting weak linear predictability in returns, even though volatility varies over time.

Likelihood-Based Estimation Results

Normal MLE (train):

$$\mu = 0.00059925, \sigma = 0.01297433$$

Log Likelihood train = 9394.88, Log likelihood test = 2332.03

AIC train = -18785.77, BIC train = -18773.62

Student-t MLE (train) with df grid search 1..150:

$$df = 3, \mu = 0.00125334, \sigma = 0.00810044$$

Log Likelihood train = 9741.49, Log likelihood test = 2385.22

AIC train = -19476.99, BIC train = -19458.76

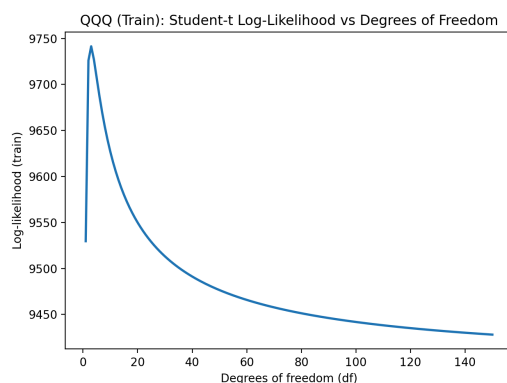
Model comparison (train):

Log Likelihood improvement (t - normal) = 346.61

Likelihood Ratio statistic = $2*(LL_t - LL_n) = 693.22$

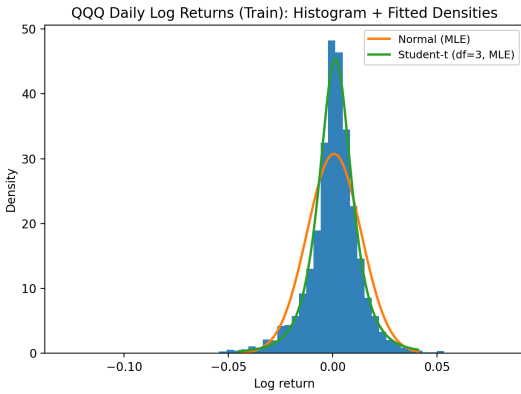
Note: Normal is the $df \rightarrow \infty$ limit of Student-t, so LR χ^2 reference is only approximate.

Maximum likelihood estimation was conducted on the training sample (2010–2022). Under the Normal model, estimated mean returns are close to zero with a constant variance parameter.



Under the Student-t specification, estimation is performed by grid-searching over degrees of freedom from 1 to 150 and maximizing the likelihood with respect to the location and scale

parameters for each candidate value. We found that log likelihood peaked at 3 degrees of freedom (left) and then subsequently fell as the degrees of freedom (v) increased.



For QQQ, the Student-t model achieves a substantially higher maximized log-likelihood than the Normal model (seen on the left). The optimal degrees of freedom parameter is estimated to be $v=3$, indicating extremely heavy tails. This improvement in fit is not merely mechanical: both AIC and BIC strongly favor the Student-t model despite its additional parameter. The log-likelihood profile over degrees of freedom further confirms that the data are far from Gaussian, with likelihood declining sharply as v increases toward the Normal limit.

limit.

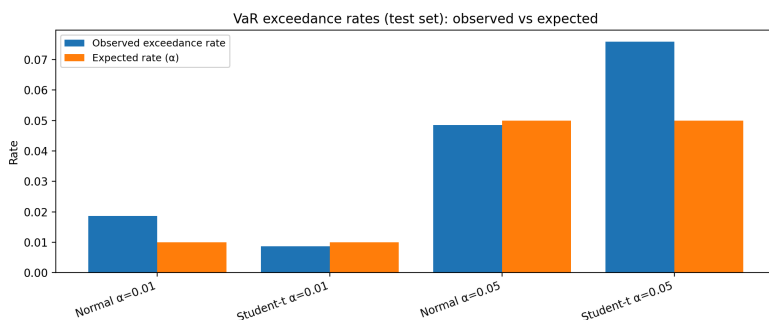
We also look at the likelihood ratio test (693.22) which is well above the threshold of 10, indicating that the t-distribution is a significantly better fit. However, this test is a bit loose for the case since the normal distribution can be thought of as the t-distribution restricted at infinite degrees of freedom. Thus violating the constraints for the likelihood test to follow a chi-squared distribution, but the ratio is so large that it is still relevant.

Out-of-sample log-likelihoods computed on the test period (2022–2025) reinforce the findings. The Student-t model continues to outperform the Normal model, suggesting that the heavy-tailed structure is not driven solely by in-sample overfitting but reflects persistent features of the return distribution.

Value at Risk: Fixed-Parameter Backtests

ticker	model	alpha	VaR_thresho ld_return	VaR_threshol d_loss	excee d_cou nt	n_t est	exceed_ rate	LR_ uc	p_ uc	LR_i nd	p_in d	LR_c c	p_cc	n00	n01	n10	n11
QQQ	Normal	0.01	-0.0295836	0.02958356	15	80 3	0.01867 995	4.867 2599 4	0.027 3709 3	1.175 2885 1	0.278 3172 5	6.042 5484 4	0.048 7390 7	773	14	14	1
QQQ	Student- t	0.01	-0.0355284	0.03552836	7	80 3	0.00871 731	0.139 4926 4	0.708 7859 2	3.980 2798	0.046 0359 2	4.119 7724 3	0.127 4684 7	789	6	6	1
QQQ	Normal	0.05	-0.0207416	0.02074163	39	80 3	0.04856 787	0.034 9908 1	0.851 6149 2	0.612 5779 4	0.433 8188 2	0.647 5687 5	0.723 4062 1	727	36	36	3
QQQ	Student- t	0.05	-0.0178099	0.01780995	61	80 3	0.07596 513	9.901 8039 4	0.001 6511 7	0.488 4031 2	0.484 6405	10.39 0207 1	0.005 5436 4	687	55	54	6

To assess the economic implications of distributional assumptions, we evaluate one-day-ahead Value at Risk (VaR) at the 1% and 5% confidence levels using parameters estimated from the training sample. VaR performance is assessed on the test sample using standard coverage tests.

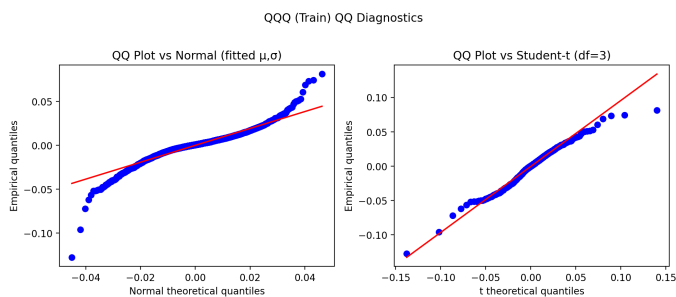


At the 1% level, the Normal model produces nearly twice as many VaR violations as expected (seen on the left), and is rejected by unconditional and conditional coverage tests. This indicates systematic underestimation of extreme downside risk. In contrast, the

Student-t model generates an exceedance frequency close to the nominal level and is not rejected by coverage tests, demonstrating substantially improved calibration in the extreme tail.

At the 5% level, the results differ. The Normal model performs adequately, with violation rates close to the expected frequency and no evidence of rejection. The Student-t model, however, generates too many violations and is rejected by coverage tests, reflecting over-conservatism at moderate quantiles. These results highlight a tradeoff: heavy-tailed models substantially improve modeling of rare, severe losses but may distort risk estimates at less extreme confidence levels.

Summary of Findings



Taken together, the results provide consistent evidence that daily equity returns for QQQ are not well described by a Gaussian distribution. Likelihood-based comparisons, information criteria, and fixed-parameter out-of-sample VaR backtests all favor a heavy-tailed Student-t specification, particularly for modeling extreme downside

risk. However, rolling-window results highlight that even heavy-tailed models do not fully resolve time-varying volatility, suggesting that distributional flexibility improves tail modeling but does not eliminate broader dynamic misspecification.

V. Conclusion and future recommendations

This paper evaluates the adequacy of the Gaussian assumption for daily equity returns using a likelihood-based comparison between Normal and Student-t models. Focusing on analysis from QQQ daily log returns, maximum likelihood estimation strongly favors the Student-t specification. The fitted degrees of freedom parameter is low ($df = 3$), indicating substantial excess kurtosis (fatter tails) relative to the Normal benchmark. Crucially, this result is reflected consistently across higher in-sample log-likelihood, lower AIC/BIC values, and improved out-of-sample likelihood performance.

Further, as seen above, several visual diagnostics reinforce these findings. Histogram overlays and QQ plots show that while both models fit the center of the return distribution reasonably well, the Normal model fails to capture extreme observations in the tails. The Student-t model substantially reduces tail deviations, particularly in the left tail, which is most relevant for downside risk. Additionally, the log-likelihood profile over degrees of freedom further confirms that QQQ returns are better characterized by heavy-tailed behavior than by a near-Gaussian distribution. In particular, out-of-sample Value-at-Risk backtesting provides additional insight into the economic implications of these distributional differences. At the 1 percent confidence level, the Normal model exhibits nearly twice the expected exceedance rate, indicating a systematic underestimation of extreme downside risk. The Student-t model, in contrast, produces exceedance rates closely aligned with the nominal level, suggesting superior calibration in the extreme tail. At the 5 percent level, however, the Normal model performs well, while the Student-t model generates a higher-than-expected violation rate, reflecting the cost of fitting very heavy tails. Taken together, these results highlight a clear tradeoff: the Student-t model improves protection against rare, severe losses but can be overly *conservative* at moderate confidence levels.

Conclusively, the evidence suggests that assuming Gaussian returns can materially understate extreme risk for equity index returns such as QQQ. While the Normal model may be adequate for modeling average behavior or moderate quantiles, it performs poorly in precisely the regions that matter most for risk management and high-stress scenarios. Decisively, the Student-t model offers a simple and effective parametric alternative that better captures empirical tail behavior.

VI. Bibliography

- Bollerslev, Tim. 1987. “A Conditionally Heteroskedastic Time Series Model for Speculative Prices and Rates of Return.” *Review of Economics and Statistics* 69 (3): 542–547.
- Cont, Rama. 2001. “Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues.” *Quantitative Finance* 1 (2): 223–236.
- Tsay, Ruey S. 2010. *Analysis of Financial Time Series*. 3rd ed. Hoboken, NJ: Wiley.
- ECON H324 Class Slides and Content

VII. Appendix

Github Repo: https://github.com/Ar4yu/Non_gaussian_market_returns

Train: 2010-01-05 → 2022-10-05 (3211 obs)

Test: 2022-10-06 → 2025-12-17 (803 obs)

Normal MLE (train):

$\mu_{\text{hat}} = 0.00059925$, $\sigma_{\text{hat}} = 0.01297433$

$LL_{\text{train}} = 9394.88$, $LL_{\text{test}} = 2332.03$

$AIC_{\text{train}} = -18785.77$, $BIC_{\text{train}} = -18773.62$

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